

CSS Bootcamp, Day 3

Stats Fundamentals: Linear Models & R/Stata Basics

22 September 2026

Note on today's schedule: Unlike other days, we are running a single, continuous block from **09:00 to 13:00**.

Our Goals for Today:

1. **Statistical Theory:** Review probability distributions, variance, hypothesis testing, and the Gauss-Markov assumptions.
2. **The OLS Estimator:** Understand the matrix formulation behind ordinary least squares.
3. **Software Interfaces:** Introduce Stata and R side-by-side for statistical modeling.
4. **Group Application:** Work collaboratively to clean data and replicate a published regression table in both languages.

Probability Distributions & Point Estimates

Statistical modeling requires us to understand the underlying distribution of a random variable, Y .

- ▶ Let Y be the true variable in the population with an unknown mean, μ_Y .
- ▶ We estimate this by drawing a random sample of n observations from the target population.
- ▶ Assume each sampled observation, y_i , is independent and identically distributed (i.i.d.):

$$y_i \stackrel{\text{iid}}{\sim} \mathcal{D}(\mu_Y, \sigma_Y^2), \quad \text{for } i = 1, \dots, n$$

We estimate the population mean (μ_Y)—the expected value—using the sample mean ($\bar{y}$):

$$\mathbb{E}[Y] = \mu_Y \quad \xrightarrow{\text{estimated by}} \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Probability Distributions & Point Estimates

Variance measures the spread of the distribution—how far data points are from the mean. We estimate the population variance (σ_Y^2) using the sample variance (s_y^2):

$$\text{Var}(Y) = \mathbb{E}[(Y - \mu_Y)^2] = \sigma_Y^2 \quad \xrightarrow{\text{estimated by}} \quad s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$$

And the square root of the variance gives us the **standard deviation**, or the average difference between y and $\hat{y}$ (s):

$$s_y = \sqrt{s_y^2}$$

Precision of our Point Estimate

How uncertain are we that $\bar{y}$ is an accurate estimate of μ_Y ?

Ideally we'd have many $\bar{y}$ estimates—each one calculated from a new random sample (with replacement) of the same population—so we could assess how uncertain we should be in any one sample mean.

But, usually, **we don't**.

- ▶ Data collection is expensive!
- ▶ We don't have all the time in the world.
- ▶ Some data are impossible to get.

Precision of our Point Estimate

We only have one sample and one same mean— $\bar{y}$. Can we still assess uncertainty?

Yes, if we make some important assumptions!

- ▶ Is our sample a random sample?
- ▶ Is each of our observations independent of the others?
- ▶ Is each of our observations from the same population?

If the answers these are “yes,” then, for sufficiently large sample sizes, we can rely on the **Central Limit Theorem** (CLT) to assess uncertainty.

Precision of our Point Estimate

The CLT states that as $n \rightarrow \infty$, repeated random samples of size n will generate a sampling distribution that is approximately normal regardless of the shape of the population distribution.

In the context of a mean specifically, this tells us that the sampling distribution of $\bar{y}$ will be normal regardless of the true shape of Y 's distribution. Further, $\bar{\bar{y}} \approx \mu_Y$!

The best way to understand the CLT is to visually simulate it. Let's do that [here](#).

(Note that the CLT holds for sufficiently large samples—say, 30 or more. But there are approximations of the normal distribution that allow for CLT-like principles, such as the t -distribution.)

Precision of our Point Estimate

Since we know how probabilities fall under a normal distribution, we only need to know a variable's population standard deviation and the sample size to calculate a measure of precision. This is the **standard error**.

$$SE(\bar{y}) = \frac{\sigma}{\sqrt{n}}$$

But we probably don't know σ , so we substitute in s to get an *estimate* of the standard error:

$$SE(\bar{y}) = \frac{s}{\sqrt{n-1}}$$

This number is the standard deviation of the sampling distribution. It tells us how far, on average, any given sample mean from samples of this size are from the true population mean.

Precision of our Point Estimate

Notice a parallel here.

Remember z-scores, which tell you how far any given observation is from the sample mean?

$$z_i = \frac{y_i - \bar{y}}{s_y}$$

You can use standard errors to do the same thing with sample means if you plug in hypothetical values for the true population parameter:

$$z_{\bar{y}_i} = \frac{\bar{y}_i - \mu_Y}{SE(\bar{y})}$$

You'll see the importance of this in a bit...

From Univariate Central Tendencies to Relationships

In social science, we're typically more interested in the point estimates for *relationships* between two or more variables.

Ordinary least squares (OLS) regression, also known simply as “linear regression,” is a standard way to do this.

We'll review bivariate OLS, where you have one dependent variable and one independent variable. Then we'll review multiple OLS, where you have one dependent variable and any number of independent/control variables.

The Gauss-Markov Assumptions

Before formulating the OLS estimator, we must ensure our data meets the Gauss-Markov assumptions. If these hold, OLS is the Best Linear Unbiased Estimator (BLUE):

1. **Linearity in Parameters:** The population model is $\mathbf{y} = \mathbf{X}\beta + \epsilon$.
2. **Random Sampling:** Observations are an unbiased, random sample from the population.
3. **No Perfect Collinearity:** The design matrix $\mathbf{X}$ has full column rank.
4. **Zero Conditional Mean:** The error term ϵ has an expected value of zero given any values of the independent variables: $\mathbb{E}[\epsilon | \mathbf{X}] = 0$.
5. **Homoskedasticity:** The variance of the errors is constant: $\text{Var}(\epsilon | \mathbf{X}) = \sigma^2 \mathbf{I}$.

The OLS Estimator: Simple Bivariate Formulation

OLS models an outcome y_i as a linear function of a single independent variable x_i , plus an error term ϵ_i :

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

OLS finds the intercept $\hat{\beta}_0$ and slope $\hat{\beta}_1$ that minimize the sum of squared residuals (SSR):

$$\text{SSR} = \sum_{i=1}^n \hat{\epsilon}_i^2 = \sum_{i=1}^n (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$

And here's how we estimate $\hat{\beta}_0$ and $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Minimizing SSR in the Bivariate Case

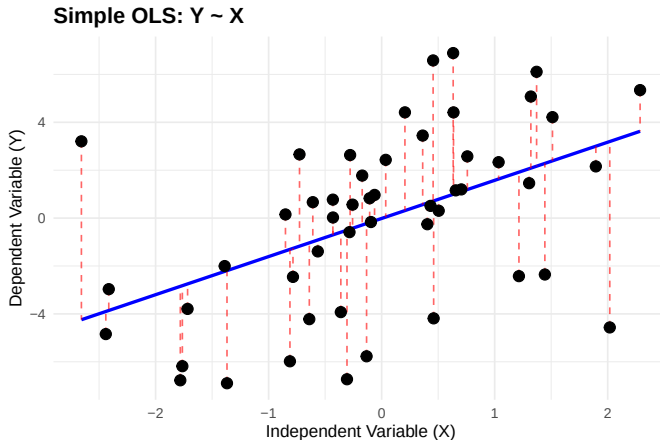


Figure: Minimizing SSR with a Straight Regression Line

Multiple OLS: The Logic of Controlling

Multiple OLS regression comes into play when you need to “isolate” the effect of an X on Y by controlling for other variables.

Let Y be our dependent variable and X our independent variable. Now let Z be an additional predictor that we want to control for.

There are four general ways X might affect Y conditional on Z :

- ▶ direct effect
- ▶ indirect effect
- ▶ spurious effect
- ▶ suppressor effect

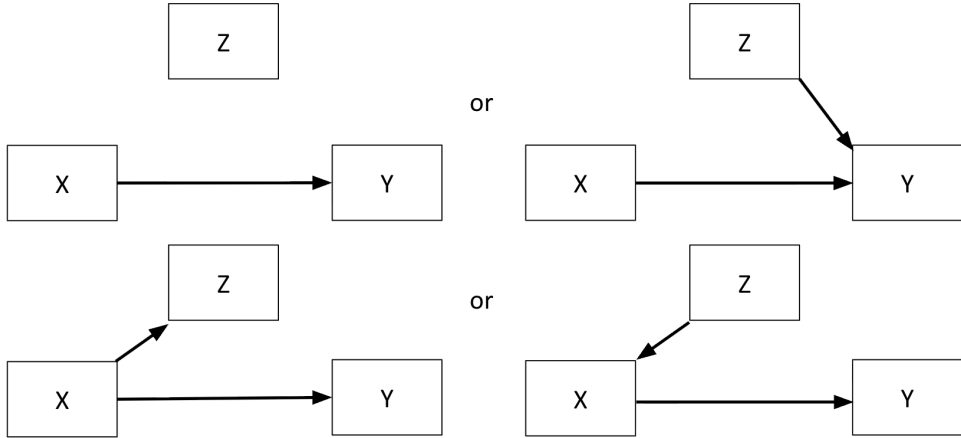


Figure: Direct Effects

Indirect Effects



Figure: Indirect Effects

(Partially) Spurious Effects

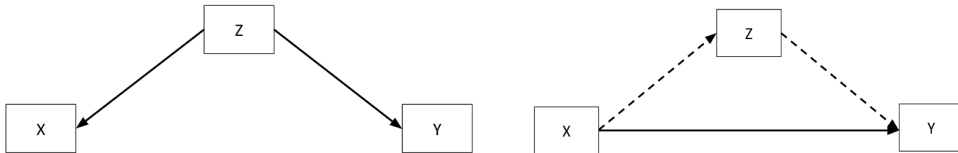


Figure: Spurious Effects and Partially Spurious Effects

Suppressor Effects

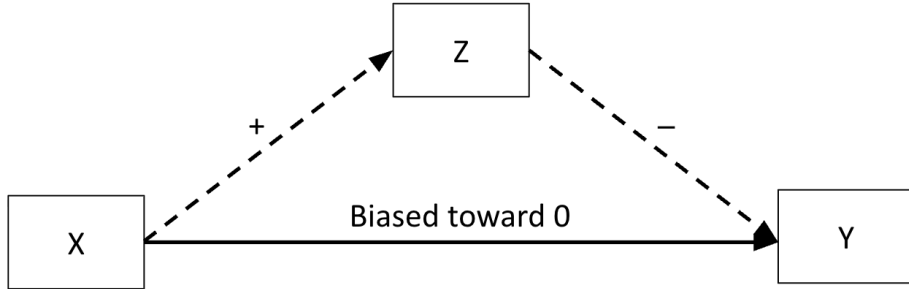


Figure: Suppressor Effects

The Logic of Controlling

When we use multiple regression, **we control for some predictors, like Z , in order to isolate any direct effect that X may have on Y . Or at least if there seems to be any direct association between them.**

So, when do we control for Z ? When we have theoretical or empirical reasons to believe that Z is correlated with *both* X and Y .

Why?

The Logic of Controlling

When Y is *only* a function of X , then ϵ (the unexplained variance) is purely random. That is:

$$\hat{\epsilon}_i = y_i - \hat{y}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)$$

But when Y is a function of both X and Z , then ϵ is *not* random—because Y is a function of more variables than we accounted for:

$$\hat{\epsilon}_i = y_i - (\hat{y}_i + \hat{\beta}_2 z_i) = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i + \hat{\beta}_2 z_i)$$

Not estimating the effect of Z in this scenario is an example of **omitted variable bias**. This can result in the effect of X being biased because its slope (β_1) is capturing the effect of X on Y **and also the portion of the effect of Z on Y that is correlated with X .**

The Logic of Controlling

We think our bivariate estimate ($\tilde{\beta}_1$) matches the multiple OLS estimate ($\hat{\beta}_1$):

$$\tilde{\beta}_1 = \hat{\beta}_1$$

But it actually mechanically includes the omitted variable's coefficient ($\hat{\beta}_2$):

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \left(\frac{\text{Cov}(x, z)}{\text{Var}(x)} \right)$$

The Logic of Controlling

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \left(\frac{\text{Cov}(x, z)}{\text{Var}(x)} \right)$$

The above shows that omitted variables will only result in bias for observed independent variables when there is a correlation between the observed variables and the omitted variable. When $\text{Cov}(X, Z) = 0$, then $\beta_1 = \hat{\beta}_1$ —regardless of whether Y is also correlated with Z .

So be smart when you're deciding on your specification of predictors!

The OLS Estimator: Matrix Formulation

Let's assume you have your mix independent and control variables.

Ordinary Least Squares (OLS) models an $n \times 1$ vector of outcomes, $\mathbf{y}$, as a linear combination of an $n \times p$ design matrix, $\mathbf{X}$, plus an $n \times 1$ error vector, ϵ :

$$\mathbf{y} = \mathbf{X}\beta + \epsilon$$

OLS finds the $p \times 1$ coefficient vector $\hat{\beta}$ that minimizes the SSR:

$$\text{SSR} = \hat{\epsilon}^T \hat{\epsilon} = (\mathbf{y} - \mathbf{X}\hat{\beta})^T (\mathbf{y} - \mathbf{X}\hat{\beta})$$

Taking the derivative with respect to $\hat{\beta}$ and setting it to zero yields the closed-form OLS solution:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

The OLS Estimator: Matrix Formulation

The y-intercept (β_0) is absorbed directly into the design matrix $\mathbf{X}$ and the coefficient vector β .

To estimate an intercept, the first column of the $n \times p$ design matrix $\mathbf{X}$ is a vector of ones (a constant). The coefficient vector β is structured so that β_0 is the first element:

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1k} \\ 1 & x_{21} & \dots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{nk} \end{bmatrix} \quad \text{and} \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}$$

Multiplying the i -th row of $\mathbf{X}$ by β recovers the standard scalar equation with the intercept:

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \epsilon_i$$

Minimizing SSR in the Multivariate Case

Multiple OLS: $Y \sim X + Z$

• Observed Data

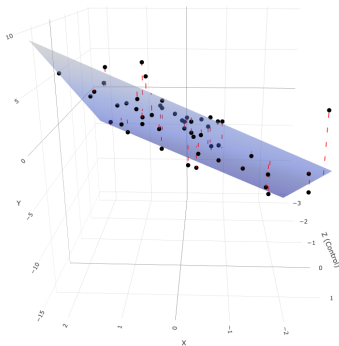


Figure: Minimizing SSR with a Regression Plane ([click here](#) for the interactive version)

Comparing Bivariate to Multiple OLS

Let's simulate some data to compare bivariate to multiple OLS when we force Y to be only a function of X and then a function of both X and Z . When does OLS adequately recover the true parameter, $\beta_1 = 2$?

Here's what we'll do with $n = 50$:

- ▶ $Z_{1i} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$
- ▶ $X_{1i} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$
- ▶ $Y_{1i} = 2X_{1i} + 3Z_{1i} + \mathcal{N}(0, 1)$
- ▶ $Z_{2i} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$
- ▶ $X_{2i} = 2Z_{2i} + \mathcal{N}(0, 1)$
- ▶ $Y_{2i} = 2X_{2i} + 3Z_{2i} + \mathcal{N}(0, 1)$

- ▶ $y_{1i} = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\epsilon}_i$
- ▶ $y_{1i} = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 z_{1i} + \hat{\epsilon}_i$
- ▶ $y_{2i} = \hat{\beta}_0 + \hat{\beta}_1 x_{2i} + \hat{\epsilon}_i$
- ▶ $y_{2i} = \hat{\beta}_0 + \hat{\beta}_1 x_{2i} + \hat{\beta}_2 z_{2i} + \hat{\epsilon}_i$

We expect a smaller difference in the $\hat{\beta}_1$ estimates when modeling y_1 than when modeling y_2 .

```
set.seed(8675309)
n <- 50
X_1 <- rnorm(n, 0, 1)
Z_1 <- rnorm(n, 0, 1) #X and Z are independent
Y_1 <- 2 * X_1 + 3 * Z_1 + rnorm(n, 0, 1) # Y depends on X & Z

Z_2 <- rnorm(n, 0, 1)
X_2 <- rnorm(n, 0, 1) + 2 * Z_2 #X depends on Z
Y_2 <- 2 * X_2 + 3 * Z_2 + rnorm(n, 0, 1) # Y depends on X & Z

df_1 <- data.frame(X_1, Z_1, Y_1,
X_2, Z_2, Y_2)

lm(Y_1 ~ X_1, data = df_1)$coefficients[2]
lm(Y_1 ~ X_1 + Z_1, data = df_1)$coefficients[2]

lm(Y_2 ~ X_2, data = df_1)$coefficients[2]
lm(Y_2 ~ X_2 + Z_2, data = df_1)$coefficients[2]
```

Comparing Bivariate to Multiple OLS

Model	$X \perp\!\!\!\perp Z$	$X \not\perp\!\!\!\perp Z$
Bivariate	1.814	3.270
Multiple	1.962	2.015

Omitted Variable Bias only occurs when the omitted variable is correlated with both the independent and dependent variables. When X and Z are correlated (and each with Y), the bivariate estimate is highly biased.

When X and Z are correlated, multiple OLS better recovers $\beta_1 = 2$. When X and Z are *not* correlated, both are pretty good.

Comparing Bivariate to Multiple OLS

We can show how the biased estimate— $\hat{\beta}_1 = 3.270$ is directly a function of both the unbiased estimate ($\hat{\beta}_1 = 2.015$, the estimate for Z ($\hat{\beta}_2 = 3.056$), and the slope from regressing z on x ($\frac{\text{Cov}(x,z)}{\text{Var}(x)}$):

```
beta1_hat <- coef(lm(Y_2 ~ X_2 + Z_2, data = df_1))["X_2"]
beta2_hat <- coef(lm(Y_2 ~ X_2 + Z_2, data = df_1))["Z_2"]

z_x <- coef(lm(Z_2 ~ X_2, data = df_1))["X_2"]

biased_estimate <- beta1_hat + (beta2_hat * z_x)
```

This gives 3.270!

Precision of $\hat{\beta}_k$: Standard Error

The **standard error**, or SE, is the standard deviation of our theoretical sampling distribution.

In the case of a regression coefficient estimate like $\hat{\beta}_{X_1}$, it is the standard deviation of the distribution we would get over a very large (potentially infinite) number of $\hat{\beta}_{X_1}$ from successive random samples of the population.

Remember that the CLT tells us that with sufficiently large sample sizes, the sampling distribution of an estimate will be approximately normal regardless of the shape of the population distribution. Further, it tells us that $\bar{\hat{\beta}}_{X_1} \approx \beta_{X_1}$.

So $SE(\hat{\beta}_{X_1})$ tells us how far, on average, any given sample estimate effect deviates from the true (yet unknown!) effect of X_1 “on” Y (controlling for any other predictors).

Precision of $\hat{\beta}_k$: Standard Error

In the case of one independent variable (predictor), X :

$$SE(\hat{\beta}) = \sqrt{\frac{SSR}{(n-2) \times \sum (X_i - \bar{X})^2}}$$

And in the case of more than one predictor, the SE for a single independent variable X_1 is:

$$SE(\hat{\beta}_{X_1}) = \sqrt{\frac{SSR}{(n-p) \times \sum (X_{i1} - \bar{X}_1)^2 \times (1 - R_{X_1}^2)}}$$

Hypothesis Testing

When we estimate a relationship in our sample, how do we know it exists in the population? We formulate competing hypotheses for a given parameter (e.g., β_{X_1}):

- ▶ **Null Hypothesis (H_0):** $\beta_{X_1} = 0$ (no effect).
- ▶ **Alternative Hypothesis (H_a):** $\beta_{X_1} \neq 0$ (an effect exists).

We evaluate our sample estimate ($\hat{\beta}_{X_1}$) using a t -statistic, scaling the estimate by its standard error (SE):

$$t = \frac{\hat{\beta}_{X_1} - 0}{\text{SE}(\hat{\beta}_{X_1})}$$

Does this look familiar? **It's the t -distribution of a z -score where we're assessing how far this sample estimate would be from a population parameter (mean) of 0!**

Hypothesis Testing

If the probability (the **p-value**) of observing a t -statistic at least this extreme is very low (typically < 0.05) under the assumption that H_0 is true, we *reject* the null hypothesis.

We call these thresholds our **α -levels**—they are the probabilities below which we are willing to reject H_0 when it was, in fact, true (a Type I error, or **false positive**). In sociology, we usually use 0.05, 0.01, and 0.001.

These values translate to rejecting H_0 when there is less than a 5%, 1%, and 0.1% chance, respectively, of observing a t -statistic at least as extreme as the one we did if the H_0 were true in the population from which the data were sampled.

When we reject H_0 , we say our result is **statistically significant**. Note that this only means that there is less than a α probability that we would observe our result if the true parameter in the population was equal to (in this case) 0!

Note that $\beta_{X_1} = 0$ and $\beta_{X_1} \neq 0$ make up a **two-tailed test**. We can also perform **one-tailed tests**.

- ▶ Right-tailed test
 - ▶ **Null Hypothesis (H_0):** $\beta_{X_1} \leq 0$ (there is a negative effect or no effect).
 - ▶ **Alternative Hypothesis (H_a):** $\beta_{X_1} > 0$ (a positive effect exists).
- ▶ Left-tailed test
 - ▶ **Null Hypothesis (H_0):** $\beta_{X_1} \geq 0$ (there is a positive effect or no effect).
 - ▶ **Alternative Hypothesis (H_a):** $\beta_{X_1} < 0$ (a negative effect exists).

Software Paradigms: Stata vs. R

Stata

- ▶ **Data format:** One dataset loaded in memory at a time.
- ▶ **Workflow:** Strongly driven by specific, concise commands for econometrics.
- ▶ **Scripts:** Uses .do files.
- ▶ *Excellent for out-of-the-box regression diagnostics and survey weights.*

R

- ▶ **Data format:** Object-oriented. You can have multiple datasets in your environment.
- ▶ **Workflow:** Highly modular, relying heavily on packages (e.g., tidyverse, modelsummary).
- ▶ **Scripts:** Uses .R files or Quarto.
- ▶ *Excellent for data wrangling, complex plots, and advanced modeling.*

Example Data

Swiss Fertility and Socioeconomic Indicators (1888) Data [1]

- ▶ Fertility and socioeconomic variables for 47 French-speaking Swiss provinces around 1888
- ▶ **Fertility Rate (DV):** I_g , a historical standardized fertility index, calculated with $\left(\frac{\sum_{i=1}^7 B_i}{\sum_{i=1}^7 M_i \times H_i} \right) \times 100$, where B_i is the number of live births to married women for 5-year age cohort i , M_i is i 's number of marriages, and H_i is the historical Hutterite fertility rate in that age cohort
- ▶ **Education (IV #1):** Percent of draftees who received education beyond primary school
- ▶ **Catholic (IV #2):** Percent of the province's population who was Catholic

Basic OLS Syntax: Stata

In Stata, running a regression and asking for summary statistics is highly concise.

```
// 1. Load the Swiss dataset directly from a public URL
import delimited "https://bit.ly/swiss-data", clear

// 2. Run regression of fertility on education and %
    Catholic
reg fertility education catholic

// 3. Request summary statistics for the variables
summarize fertility education catholic
```

Notice that we do not have to tell Stata *which* dataset to look at in the regress command, because only one dataset is in memory at a time.

Basic OLS Syntax: R

In R, we must assign our regression to an object and explicitly call our data.

```
# 1. Load a built-in dataset (Swiss demographics, 1888)
data(swiss)

# 2. Run the regression and save it as an object
model <- lm(Fertility ~ Education + Catholic,
            data = swiss)

# 3. Print the results table
summary(model)
```

Because R is object-oriented, you can extract specific elements from the `model` object later, such as residuals via `resid(model)`.

```
Call:
lm(formula = Fertility ~ Education + Catholic, data = swiss)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.042	-6.578	-1.431	6.122	14.322

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	74.23369	2.35197	31.562	< 2e-16	***
Education	-0.78833	0.12929	-6.097	2.43e-07	***
Catholic	0.11092	0.02981	3.721	0.00056	***

Signif. codes:	0	***	0.001	**	0.01	*	0.05	.
	0.1		1					

Residual standard error: 8.331 on 44 degrees of freedom

Multiple R-squared: 0.5745, Adjusted R-squared: 0.5552

F-statistic: 29.7 on 2 and 44 DF, p-value: 6.849e-09

Group Application: The Replication Challenge

Follow the instructions here: bit.ly/css_bootcamp_day2.

References I

- [1] Frederick Mosteller and John W Tukey. “Data analysis and regression. A second course in statistics”. In: *Addison-Wesley series in behavioral science: quantitative methods* (1977).



END OF SESSION



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